

Financial Econometrics: Maximum Likelihood Estimation

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Note: This lecture note covers Ch 1, 2, 3 (The Maximum Likelihood Principle, Properties of Maximum Likelihood Estimators, Numerical Estimation Methods) of Econometric Modelling with Time Series: Specification, Estimation and Testing by V. L. Martin, A. S. Hurn and D. Harris.

Maximum Likelihood Estimation

MLE for time series data

Example: MLE for ARMA(p,q) model

Review

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What is the maximum likelihood estimation?

- ▶ It is one of many estimation methods like the Ordinary Least Squares (OLS) estimation.
- ▶ Depending on which objective function is used in estimation, we have a different estimation method. What does it mean? Let's see the next slide for example!

What is the maximum likelihood estimation?

$$\text{Model: } y_i = \mu + e_i, \\ e_i \sim \text{i.i.d.} N(0, \sigma^2), \quad i = 1, 2, \dots, N.$$

- ▶ Ordinary Least Squares Estimator (OLS: 최소 자승법)

$$\text{Sample Equation: } y_i = \hat{\mu} + \hat{e}_i, \quad i = 1, 2, \dots, N$$

$\hat{\mu}$: our estimate (추정치) for μ ; $\hat{e}_i$: our estimate for e_i ; $\hat{e}_i$ is called a residual (잔차항).

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N y_i$$

- ▶ Objective function of OLS (Sum of Squared Residuals, SSR):

$$\min \sum_{i=1}^N \hat{e}_i^2 = \min \sum_{t=1}^T (y_i - \hat{\mu})^2 \quad \text{w.r.t } \hat{\mu}$$

What is the maximum likelihood estimation?

$$\text{Model: } y_i = \mu + e_i$$

$$e_i \sim i.i.d.N(0, \sigma^2), \quad i = 1, 2, \dots, N.$$

- ▶ The OLS estimator of μ is given by

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N y_i$$

$$E(\hat{\sigma}^2) = \sigma^2$$

- ▶ The unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^N \hat{e}_i^2 = \frac{1}{N-1} \sum_{i=1}^N (y_i - \hat{\mu})^2$$

Maximum likelihood estimation for?

Given Data : $y_1, y_2, \dots, y_n$ $\Rightarrow$ unknown μ, σ^2

- ▶ The maximum likelihood estimation is referred to as MLE in short.
- ▶ For MLE, we use an objective function that gives the likelihood of sample data.
- ▶ An MLE estimate (ML 추정치) is the parameter value that maximizes the likelihood of the sample data.

Maximum likelihood estimation

Definition

The Likelihood value is the value that represents how much likely sample data are generated from the population (모집단) with particular characteristics. These characteristics are defined by model parameters.

Maximum likelihood estimation

$$\text{Model: } y_i = \mu + e_i$$

$$e_i \sim i.i.d.N(0, \sigma^2), \quad i = 1, 2, \dots, N.$$

Example

Sample data: $y_i \in \{2, 7, 5, 10, 6\}$

Parameter values: $\{\mu = 6, \sigma^2 = 9\}$ v.s. $\{\mu = 30, \sigma^2 = 25\}$

- ▶ Which parameter values are more likely to generate the observed data?
- ▶ Likelihood function: $L(\mu, \sigma^2)$
- ▶ In other words, which likelihood value is bigger?
 $L(\mu = 6, \sigma^2 = 8)$ or $L(\mu = 30, \sigma^2 = 15)$?

Maximum likelihood estimation

<pdf>

	pdf	LF
ex-ante	μ, σ^2	$y_1 \dots y_n$
ex-post	y_1 r.v	μ, σ^2

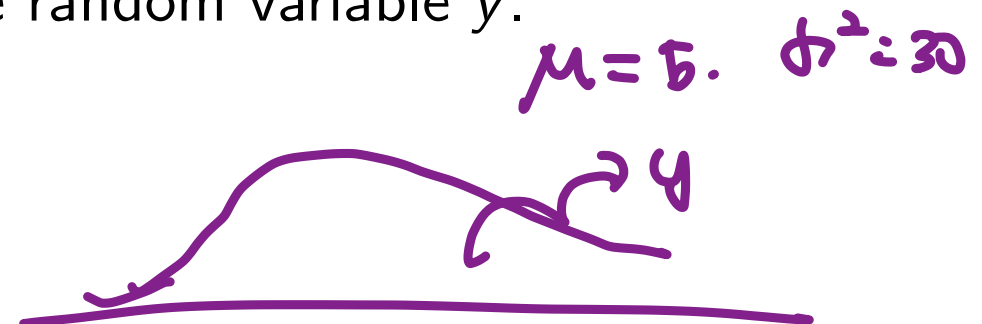
- Suppose that we draw one sample (y) from the population $N(\mu, \sigma^2)$ and the true values of μ and σ^2 are already known.

$$f(y; \theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2} (y - \mu)^2\right)$$

$\theta = \{\mu, \sigma^2\} : \text{known}$

p.d.f

- Before sampling, we don't know y . It is a random variable.
- If the values of μ and σ^2 are already given, we can make probabilistic prediction for the random variable y .



Probability density function

► $Pr(y > \mu) = \int_{\mu}^{\infty} f(y; \theta) dy = 0.5$



► $Pr(10 > y > 4) = \int_4^{10} f(y; \theta) dy$



- We can use the Probability Density Function (PDF) for probabilistic prediction.

Probability density function

- ▶ What if we draw independent samples $(y_1, y_2, \dots, y_N)$ from the same population $N(\mu, \sigma^2)$? ✓
- ▶ What is $Pr(y_1 > 10, y_2 > 0.3, \dots, y_N < -10)$?
- ▶ We need the Joint Probability Density Function (JPDF) for this case to make probabilistic prediction.
- ▶ JPDF of $\{y_1, y_2, \dots, y_N\} = f(y_1, y_2, \dots, y_N, \theta)$ where $\theta = \{\mu, \sigma^2\}$.
JPDF!
- ▶ Marginal PDF of $y_i = f(y_i; \theta)$

||

$L(\theta | y_1, \dots, y_N)$

Probability density function

- ▶ Note: If two random variables, A and B, are independent, $f(A, B) = f(A)f(B)$.

- ▶ $f(\mathbf{Y}; \theta) = f(y_1, y_2, \dots, y_N; \theta) = \prod_{i=1}^N f(y_i; \theta)$ where
 $\mathbf{Y} = [y_1, y_2, \dots, y_N]'$
 $= f(y_1; \theta) f(y_2; \theta) \dots$

- ▶ If $y_i \sim i.i.d.N(\mu, \sigma^2)$, the (joint) pdf of $\mathbf{Y}$ is given by

$$f(\mathbf{Y}; \theta) = (2\pi\sigma^2)^{-N/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mu)^2 \right)$$

$$\theta = \{\mu, \sigma^2\}$$

Likelihood function

- ▶ So far, we assume that $\{\mu, \sigma^2\}$ are known. But in reality, they are unknown model parameters.
- ▶ Although we don't know $\{\mu, \sigma^2\}$, we have the sample data $\mathbf{Y} = [y_1, y_2, \dots, y_N]'$ from the population.
- ▶ How do we learn about unknown $\{\mu, \sigma^2\}$ from the information in the sample data?
- ▶ Likelihood function:

$$L(\mu, \sigma^2; y_1, y_2, \dots, y_N) = f(y_1, y_2, \dots, y_N; \mu, \sigma^2)$$

- ▶ The likelihood function is basically the joint probability density function. But the likelihood function treats $\{\mu, \sigma^2\}$ as unknowns and $\mathbf{Y} = [y_1, y_2, \dots, y_N]'$ as the given information.

Maximum Likelihood Estimation

next..

- Objective function of MLE:

$$\max L(\mu, \sigma^2; \mathbf{Y}) \quad \text{w.r.t.} \quad \mu, \sigma^2$$

Joint pdf (red arrow pointing to L)
Given (blue arrow pointing to $\mathbf{Y}$)
EQ (red arrow pointing to the equation)

- For a monotonic transformation, the solutions to the two maximization problems are identical.

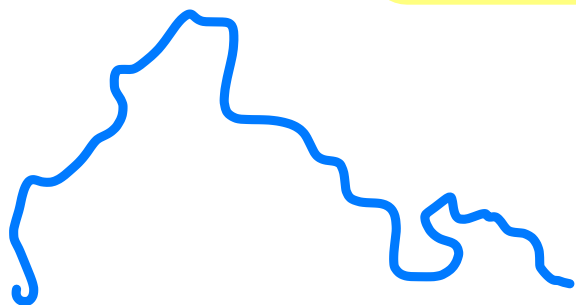
$$\text{e.g., } \arg \max_x (x - 5)^2 = \arg \max_x \ln((x - 5)^2)$$

(blue arrow pointing to $\ln$)

- Accordingly, the objective function of MLE can be expressed as

$$\max \ln L(\mu, \sigma^2; \mathbf{Y}) \quad \text{w.r.t.} \quad \mu, \sigma^2$$

$\Rightarrow$ numerical optimization



Maximum Likelihood Estimation

• closed-form μ , σ^2 solve가 가능함

- First order conditions for MLE

$$\frac{\partial \ln L(\mu, \sigma^2; \mathbf{Y})}{\partial \mu} = 0, \quad \frac{\partial \ln L(\mu, \sigma^2; \mathbf{Y})}{\partial \sigma^2} = 0$$

- ML estimates $\hat{\mu}$ and $\hat{\sigma}^2$ are the values of $\{\mu, \sigma^2\}$ that satisfy the above two first order conditions.

* < MLE : example >

$$y_i = \underbrace{X_i'}_{1 \times k} \underbrace{\beta}_{k \times 1} + e_i, \quad 1 \times 1$$

$$e_i \sim \text{Normal} \quad \textcircled{1}$$

fixed.

$$y_i \sim \text{Normal}$$

- Step 1: $y_i \sim ?$

by ①) $E(\cdot)$
 $\text{Var}(\cdot)$

$$\begin{cases} E(e_i) = 0 \quad \forall i \\ \text{Var}(e_i) = \sigma^2 \quad \forall i \end{cases}$$

$$y_i \sim N(X_i' \beta, \sigma^2)$$

- Step 2: $f(y_i)$? (y_i pdf?)

$$f(y_i; \underline{\beta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ - \frac{(y_i - X_i' \beta)^2}{\sigma^2} \right\}$$

- Step 3: $y_1, y_2, \dots, y_n$ joint pdf?

$$f(y_1, y_2, y_3, \dots, y_n; \beta, \sigma^2) = \prod_{i=1}^n f(y_i; \beta, \sigma^2)$$

$$e_i \sim \text{iid}$$

$$y_1, \dots, y_n \sim \text{joint pdf}$$

$$= (2\pi\sigma^2)^{-\frac{n}{2}} \exp \left\{ -\frac{\sum_{i=1}^n (y_i - x_i'\beta)^2}{2\sigma^2} \right\} = L_0 F = L(\beta, \sigma^2; y_1, \dots, y_n)$$

- Step 4: $L \Rightarrow \ln L$

$$\ln L(\beta, \sigma^2; y_1, \dots, y_n) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{\sum_{i=1}^n (y_i - x_i'\beta)^2}{2\sigma^2}$$

$\Downarrow$

Step 5: $\frac{\partial \ln L}{\partial \beta} = 0, \quad \frac{\partial \ln L}{\partial \sigma^2} = 0.$

$\nearrow$

$\hat{\beta}_{ML}$ $\hat{\sigma}_{ML}^2$

$\therefore \hat{\beta}_{ML} = \hat{\beta}_{OLS},$

$\hat{\sigma}_{ML}^2 = \frac{1}{n} \sum \hat{e}_i^2$

Maximum Likelihood Estimation: Generalization

- ▶ Suppose we have a random sample from the pdf $f(y_i; \theta)$ and we are interested in estimating θ .
- ▶ The maximum likelihood estimator, denoted $\hat{\theta}_{mle}$, is the value of θ that maximizes $L(\theta | \mathbf{Y})$. That is,

$$\hat{\theta}_{mle} = \arg \max_{\theta} L(\theta | \mathbf{Y})$$

- ▶ It is often quite difficult to directly maximize $L(\theta | \mathbf{Y})$. Since $\ln(\cdot)$ is a monotonic function, we have

$$\hat{\theta}_{mle} = \arg \max_{\theta} \ln L(\theta | \mathbf{Y})$$

Maximum Likelihood Estimation: Generalization

With **random sampling**, the log-likelihood has the particularly simple form

$$\ln L(\theta \mid \mathbf{Y}) = \ln \left(\prod_{i=1}^n f(y_i; \theta) \right) = \sum_{i=1}^n \ln f(y_i; \theta)$$

Maximum Likelihood Estimation: Generalization

↙ $\{y_1 \dots y_n\} = \mathbf{Y}$

- ▶ We find the MLE by differentiating $\ln L(\theta \mid \mathbf{Y})$ and solving the first order conditions

$$\frac{\partial \ln L(\hat{\theta}_{mle} \mid \mathbf{Y})}{\partial \theta} = \mathbf{0}$$

- ▶ If we have p unknown model parameters, the model parameter set θ is $(p \times 1)$ vector. Then, the first order conditions define p potentially non-linear equations in p unknown values:

$$\frac{\partial \ln L(\hat{\theta}_{mle} \mid \mathbf{Y})}{\partial \theta} = \begin{pmatrix} \frac{\partial \ln L(\hat{\theta}_{mle} \mid \mathbf{Y})}{\partial \theta_1} \\ \vdots \\ \frac{\partial \ln L(\hat{\theta}_{mle} \mid \mathbf{Y})}{\partial \theta_k} \end{pmatrix} = \mathbf{0} \quad \swarrow$$

Maximum Likelihood Estimation: Generalization

- ▶ The ~~vector of derivatives of the log likelihood function~~ is called the **score vector** and is denoted

$$S(\theta \mid \mathbf{Y}) = \frac{\partial \ln L(\theta \mid \mathbf{Y})}{\partial \theta}$$

$(p \times 1)$

- ▶ By definition, the MLE satisfies


$$S(\hat{\theta}_{mle} \mid \mathbf{Y}) = \mathbf{0}$$

- ▶ Under random sampling

$$S(\theta \mid \mathbf{Y}) = \sum_{i=1}^n \frac{\partial \ln f(y_i; \theta)}{\partial \theta} = \sum_{i=1}^n S(\theta \mid y_i)$$

where

$$S(\theta \mid y_i) = \frac{\partial \ln f(y_i; \theta)}{\partial \theta}$$

Derivation for MLE: Linear Regression Model

Consider a multiple linear regression model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i} + \epsilon_i, \quad \epsilon_i \sim i.i.d.N(0, \sigma^2)$$

for $i = 1, 2, \dots, N$.

- ▶ Our goal is to estimate $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]'$ and σ^2 given data $\mathbf{y} = [y_1, y_2, \dots, y_N]'$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]'$ where $\mathbf{x}_i = [1, x_{1,i}, x_{2,i}, \dots, x_{k,i}]'$.
- ▶ The model parameter set is $\theta = \{\boldsymbol{\beta}, \sigma^2\}$.
- ▶ The matrix form of the model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{E}$$

where $\mathbf{E} = [\epsilon_1, \epsilon_2, \dots, \epsilon_N]'$.

Derivation for MLE: Linear Regression Model

- The log-likelihood function for the linear regression model is

$$\ln L(\theta \mid \mathbf{y}, \mathbf{X}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$


$$\begin{array}{l} \underline{e'e} = \sum e_i^2 \\ \underline{1 \times n, n \times 1} \end{array}$$

Derivation for MLE: Linear Regression Model

- ▶ The Maximum Likelihood Estimator (MLE) of θ satisfies $S(\hat{\theta}_{mle} \mid \mathbf{y}, \mathbf{X}) = 0$ where $S(\theta \mid \mathbf{y}, \mathbf{X}) = \frac{\partial}{\partial \theta} \ln L(\theta \mid \mathbf{y}, \mathbf{X})$ is the score vector.
- ▶ Now

$$\begin{aligned}\frac{\partial \ln L(\theta \mid \mathbf{y}, \mathbf{X})}{\partial \boldsymbol{\beta}} &= \frac{-1}{2\sigma^2} \frac{\partial}{\partial \boldsymbol{\beta}} [\mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}] \\ &= -(\sigma^2)^{-1} [-\mathbf{X}'\mathbf{y} + \mathbf{X}'\mathbf{X}\boldsymbol{\beta}] = 0 \\ \frac{\partial \ln L(\theta \mid \mathbf{y}, \mathbf{X})}{\partial \sigma^2} &= -\frac{n}{2} (\sigma^2)^{-1} \\ &\quad + \frac{1}{2} (\sigma^2)^{-2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\end{aligned}$$

Derivation for MLE: Linear Regression Model

- Solving $\frac{\partial \ln L(\hat{\theta}_{mle} | \mathbf{y}, \mathbf{X})}{\partial \beta} = 0$ for $\hat{\beta}_{mle}$ gives

$$\hat{\beta}_{mle} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y} = \hat{\beta}_{OLS}$$

- Next, solving $\frac{\partial \ln L(\hat{\theta}_{mle} | \mathbf{y}, \mathbf{X})}{\partial \sigma^2} = 0$ for $\hat{\sigma}_{mle}^2$ gives

$$\hat{\sigma}_{mle}^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\beta}_{mle})' (\mathbf{y} - \mathbf{X}\hat{\beta}_{mle})$$

$$\hat{\sigma}_{mle}^2 \neq \hat{\sigma}_{OLS}^2 = \frac{1}{n-k} (\mathbf{y} - \mathbf{X}\hat{\beta}_{OLS})' (\mathbf{y} - \mathbf{X}\hat{\beta}_{OLS})$$



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MLE for time series data

- ▶ For **iid time series data** with marginal pdf $f(y_t; \theta)$, the joint density function for a sample $\mathbf{y} = (y_1, \dots, y_T)$ is simply the product of the marginal densities for each observation

$$\underline{f(\mathbf{y}; \theta)} = f(y_1, \dots, y_T; \theta) = \prod_{t=1}^T \underline{f(y_t; \theta)}$$

- ▶ The likelihood function is this joint density treated as a function of the parameters θ given the data $\mathbf{y}$:

$$L(\theta | \mathbf{y}) = L(\theta | y_1, \dots, y_T) = \prod_{t=1}^T f(y_t; \theta)$$

- ▶ Then, the log-likelihood is given in a simple form

$$\ln L(\theta | \mathbf{y}) = \sum_{t=1}^T \ln f(y_t; \theta)$$

MLE for time series data

$\rightarrow e_t \sim \text{i.i.d.}$ (X)
 $\rightarrow e_t$... **seriously correlated.**

- For a sample from a covariance stationary time series $\{y_t\}$, the construction of the log-likelihood given above typically doesn't work because the random variables in the sample $(y_1, \dots, y_T)$ are not i.i.d. in most cases.

- **Note:** If we have two random variables, A and B, the joint probability can be expressed by the product of the marginal probability and the conditional probability as

$$p(A, B) = p(B|A)p(A)$$

- **Note:** If we have three random variables, A, B, and C, the joint probability can be expressed as

$$p(A, B, C) = p(C|A, B)p(B|A)p(A)$$

$\Rightarrow p(A, B, C, D)$

MLE for time series data

- ▶ Typically, for MLE, we employ factorization of the joint density into a series of conditional densities and the density of a set of initial values.
- ▶ Consider the joint density of two adjacent observations $f(y_2, y_1; \theta)$ from a covariance stationary time series. The joint density can always be factored as the product of the conditional density of y_2 given y_1 and the marginal density of y_1 :

$$f(y_2, y_1; \theta) = f(y_2 | y_1; \theta) f(y_1; \theta)$$

- ▶ For three observations, the factorization becomes

$$f(y_3, y_2, y_1; \theta) = f(y_3 | \underbrace{y_2, y_1}_{\text{past}}; \theta) f(y_2 | y_1; \theta) f(y_1; \theta)$$

MLE for time series data

$y_1 \dots y_T$

$$\Rightarrow f(y_t | I_{t-1})$$

- In general, the conditional marginal factorization has the form

$$f(\underline{y_T}, \dots, \underline{y_1}; \theta) = \left(\prod_{t=p+1}^T f(y_t | I_{t-1}, \theta) \right) \cdot f(y_p, \dots, y_1; \theta)$$

Handwritten notes:
- A pink circle highlights the product term.
- A red arrow points from the text "Tide dependence Data set size!" to the product term.
- Blue lines underline the terms y_t and I_{t-1} in the product term.

where $I_t = \{y_t, \dots, y_1\}$ denotes the information available at time t , and $y_p, \dots, y_1$ denotes the p initial observations.

- The log-likelihood function may then be expressed as

$$\ln L(\theta | \underline{y}) = \sum_{t=p+1}^T \ln f(y_t | I_{t-1}, \theta) + \ln f(y_p, \dots, y_1; \theta)$$

Handwritten notes:
- A black circle highlights the $\underline{y}$ in the log-likelihood function.
- A black wavy line is drawn below the equation.

MLE for time series data

- ▶ In the maximum likelihood estimation of time series models, two types of maximum likelihood estimates may be computed.
- ▶ The first type is based on maximizing the conditional log-likelihood function. These estimates are called conditional MLEs and are defined by

$$\hat{\theta}_{cmle} = \arg \max_{\theta} \sum_{t=p+1}^T \ln f(y_t \mid I_{t-1}, \theta)$$

- ▶ The second type is based on maximizing the exact log-likelihood function. These estimates are called exact MLEs, and are defined by

$$\hat{\theta}_{mle} = \arg \max_{\theta} \sum_{t=p+1}^T \ln f(y_t \mid I_{t-1}, \theta) + \ln f(y_p, \dots, y_1; \theta)$$

MLE for AR(1) model

- ▶ Consider the stationary AR(1) model

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \varepsilon_t \sim i.i.d.N(0, \sigma^2), t = 1, \dots, T$$
$$\theta = \{c, \phi, \sigma^2\}', |\phi| < 1$$

- ▶ Conditional on I_{t-1}

$$y_t \mid I_{t-1} \sim N(c + \phi y_{t-1}, \sigma^2), t = 2, \dots, T$$

which only depends on y_{t-1} .

- ▶ The conditional density $f(y_t \mid I_{t-1}, \theta)$ is then

$$f(y_t \mid y_{t-1}, \theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2} (y_t - c - \phi y_{t-1})^2\right),$$

for $t = 2, 3, \dots, T$.

MLE for AR(1) model

- To determine the marginal density for the initial value y_1 , recall that for a stationary AR(1) process

$$E[y_1] = \mu = \frac{c}{1 - \phi}$$
$$\text{var}(y_1) = \frac{\sigma^2}{1 - \phi^2}$$

- It follows that

$$y_1 \sim N\left(\frac{c}{1 - \phi}, \frac{\sigma^2}{1 - \phi^2}\right)$$
$$f(y_1; \theta) = \left(2\pi \frac{\sigma^2}{1 - \phi^2}\right)^{-1/2} \exp\left(-\frac{1 - \phi^2}{2\sigma^2} \left(y_1 - \frac{c}{1 - \phi}\right)^2\right)$$

MLE for AR(1) model

- ▶ The conditional log-likelihood function for $y_2, y_3, \dots, y_T$ is

$$\begin{aligned} \sum_{t=2}^T \ln f(y_t \mid y_{t-1}, \boldsymbol{\theta}) &= \frac{-(T-1)}{2} \ln(2\pi) - \frac{(T-1)}{2} \ln(\sigma^2) \\ &\quad - \frac{1}{2\sigma^2} \sum_{t=2}^T (y_t - c - \phi y_{t-1})^2 \end{aligned}$$

- ▶ The marginal log-likelihood for the initial value y_1 is

$$\ln f(y_1; \boldsymbol{\theta}) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \left(\frac{\sigma^2}{1 - \phi^2} \right) - \frac{1 - \phi^2}{2\sigma^2} \left(y_1 - \frac{c}{1 - \phi} \right)^2$$

MLE for AR(1) model

- ▶ Notice that the conditional log-likelihood function has the form of the log-likelihood function for a linear regression model with normal errors.
- ▶ It follows that the conditional MLEs for c and ϕ are identical to the least squares estimates from the regression

$$y_t = c + \phi y_{t-1} + \varepsilon_t, t = 2, \dots, T$$

MLE for AR(1) model

- ▶ The exact log-likelihood function is a **non-linear function of the parameters θ** , and so there is no closed form solution for the exact mles.
- ▶ The exact mles must be determined by numerically maximizing the exact log-likelihood function.

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MLE for ARMA(p,q) model

Consider MLE of the following ARMA(p,q) model:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} \\ + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \\ \varepsilon_t \sim \text{i.i.d.} N(0, \sigma^2)$$

unobserved ✓
component ✓
(VC
model)

- Step 1:** Find conditional mean $\mu_{t|t-1} = E[y_t | I_{t-1}]$, conditional variance $V_{t|t-1} = \text{Var}[y_t | I_{t-1}]$, unconditional mean $\mu = E[y_t]$ and unconditional variance $V = \text{Var}[y_t]$.
- Step 2:** Construct the likelihood function based on the joint density of $y_1, y_2, \dots, y_T$.

$$L(\boldsymbol{\theta} | \mathbf{y}) = \sum_{t=1}^T f(y_t | I_{t-1}, \boldsymbol{\theta}), \quad y_t | I_{t-1} \sim N(\mu_{t|t-1}, V_{t|t-1})$$

- Step 3:** Maximize the log likelihood function w.r.t $\boldsymbol{\theta}$ using a numerical optimization algorithm like the Newton-Raphson algorithm.

MLE for ARMA(p,q) model

- ▶ Note that I_0 is the information set at time 0 when we don't have any observation yet. Thus, $\mu_{1|0} = \mu$ and $V_{1|0} = V$.

$$y_1 | I_0 \sim N(\mu, V)$$

- ▶ We can compute the unconditional moments (μ and V) easily using the matrix representation of the ARMA(p,q) model.

How about $\mu_{2|1}$, $V_{2|1}$, $\mu_{3|2}$ and $V_{3|2}$, ... so on?

- ▶ Later, we will learn (linear) state-space models and the Kalman filter. These tools allow us to easily compute all the conditional moments of the ARMA(p,q) model, which makes MLE easy to implement in practice.

→ kalman filter.

Hamilton filter

Soon !!

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Statistical inference and prediction with ARMA(p,q)

For a general ARMA model, we can do impulse response analysis and prediction.

Step 1: Use MLE to estimate all model parameters.

Step 2: Construct the matrix representation of your ARMA model.

Step 3: Given the coefficient matrix $\mathbf{F}$, perform impulse response analysis and prediction.

Estimation of ARMA in Python

The following Python code demonstrates how to estimate an ARMA(2,2) model given time series data **y**:

```
1 # import necessary packages and libraries
2 import numpy as np
3 import pandas as pd
4 import statsmodels.api as sm
5 from statsmodels.tsa.arima.model import ARIMA
6
7 # Fit ARMA model - here we fit an ARMA(2,2) model
8 # You need to choose the order based on your actual
   data
9 model = ARIMA(y, order=(2, 0, 2))
10 results = model.fit()
11
12 # Perform out-of-sample prediction
13 # 'start' and 'end' can be modified to predict
   different out-of-sample time points
14 n_forecast = 10 # number of steps to forecast
15 forecast = results.get_forecast(steps=n_forecast)
16 predicted_mean = forecast.predicted_mean
17 conf_int = forecast.conf_int()
```

The End. Next, we will learn unit root tests to statistically check if a time series variable is stationary or non-stationary.

Also, we will see how this univariate prediction model is applied to obtain expected returns of stocks.

Appendix: Newton-Raphson Iteration

Goal: Solve the following using an iterative scheme:

$$\hat{\theta} = \arg \max_{\theta} \ln L(\theta \mid \mathbf{x})$$

- Idea: Consider 2nd order Taylor Series Expansion (TSE) of $\ln L(\theta \mid \mathbf{x})$ about starting value $\hat{\theta}_1$

$$\begin{aligned} \ln L(\theta \mid \mathbf{x}) = & \ln L(\hat{\theta}_1 \mid \mathbf{x}) + \boxed{\frac{\partial \ln L(\hat{\theta}_1 \mid \mathbf{x})}{\partial \theta'}} (\theta - \hat{\theta}_1) \\ & + \frac{1}{2} (\theta - \hat{\theta}_1)' \frac{\partial^2 \ln L(\hat{\theta}_1 \mid \mathbf{x})}{\partial \theta \partial \theta'} (\theta - \hat{\theta}_1) + \text{error} \end{aligned}$$

Appendix: Newton-Raphson Iteration

- ▶ 'Score matrix' (the first derivative of the log likelihood function) denoted by S is given as:

$$S = \frac{\partial \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta'}$$

- ▶ 'Hessian matrix' (the second derivative of the log likelihood function) denoted by H is given as:

$$H = \frac{\partial^2 \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta \partial \theta'}$$

Information matrix
also called

Appendix: Newton-Raphson Iteration

- Now maximize 2nd order TSE wrt θ . The FOCs are

$$\mathbf{0}_{p \times 1} = \frac{\partial \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta} + \frac{\partial^2 \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta \partial \theta'} (\hat{\theta}_2 - \hat{\theta}_1)$$

- Solve for $\hat{\theta}_2$

$$\begin{aligned} \hat{\theta}_2 &= \hat{\theta}_1 - \left[\frac{\partial^2 \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta \partial \theta'} \right]^{-1} \frac{\partial \ln L(\hat{\theta}_1 | \mathbf{x})}{\partial \theta} \\ &= \hat{\theta}_1 - H(\hat{\theta}_1 | \mathbf{x})^{-1} S(\hat{\theta}_1 | \mathbf{x}) \end{aligned}$$

- This suggests the iterative scheme

$$\hat{\theta}_{n+1} = \hat{\theta}_n - H(\hat{\theta}_n | \mathbf{x})^{-1} S(\hat{\theta}_n | \mathbf{x})$$

- Iteration stops when

$$S(\hat{\theta}_n | \mathbf{x}) \approx 0$$

Appendix: Newton-Raphson Iteration

- ▶ A Newton-Raphson type algorithm is used for the maximization which leads to the iterative scheme

$$\hat{\boldsymbol{\theta}}_{mle,n} = \hat{\boldsymbol{\theta}}_{mle,n-1} - \mathbf{H}(\hat{\boldsymbol{\theta}}_{mle,n-1})^{-1} \mathbf{S}(\hat{\boldsymbol{\theta}}_{mle,n-1})$$

where $\mathbf{H}(\hat{\boldsymbol{\theta}})$ is an estimate of the Hessian matrix (2nd derivative of the log-likelihood function), and $\mathbf{S}(\hat{\boldsymbol{\theta}})$ is an estimate of the score vector (1st derivative of the loglikelihood function).

< Numerical optimization > closed-form solution, exact.

Goal:

$$\frac{\partial \ln L(\theta)}{\partial \theta} = 0$$

$\theta: \theta_0, \theta_1$

→ f.o.c, = 0,
by $\hat{\theta}_{MLE}$.

- (note) $f(z) \approx$ $z = z^*$, Taylor series expansion.
↳ continuous, 미분가능.

$$f(z) = \underline{f(z^*)} + f'(z^*)(z - z^*)$$

↓
다항식
근사

$$+ \frac{1}{2!} f''(z^*)(z - z^*)^2 + \dots$$

+ ...

→ "근사화"
(linear approximation)

$$\theta = \begin{matrix} 3 \times 1 \\ \begin{bmatrix} \beta_1 \\ \beta_2 \\ \sigma^2 \end{bmatrix} \end{matrix}$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} \approx \underbrace{\frac{\partial \ln L(\theta_j)}{\partial \theta} + \frac{\partial^2 \ln L(\theta_j)}{\partial \theta \partial \theta'} (\theta - \theta_j) = 0}_{\theta = ?}$$

$$\theta = \theta_j$$

$$\theta - \theta_j = - \left[\frac{\partial^2 \ln L(\theta_j)}{\partial \theta \partial \theta'} \right]^{-1} \frac{\partial \ln L(\theta_j)}{\partial \theta}$$

S.O.C < 0 f.o.c

$$\theta_{j+1} = \theta_j - \left[\frac{\partial^2 \ln L(\theta_j)}{\partial \theta \partial \theta'} \right]^{-1} \frac{\partial \ln L(\theta_j)}{\partial \theta}$$

↘ ↗ ↘


$$\theta_{j+1} = \theta_j - \left[\right]^{-1} \frac{2 \ln(\theta_j)}{2\theta}$$

S.O.C

$$\theta_1 = \theta_0 - \underbrace{\ast(\theta)}$$

$$\theta_{j+1} = \theta_j - \alpha (\theta_j)$$

$$\frac{\partial \ln L(\theta_{j+1})}{\partial \theta} \approx 0,$$

$$\theta_{j+1} \geq \theta_j$$

$$-E \left[\frac{\partial^2 \ln L(\theta)}{\partial \theta \partial \theta'} \right] = I(\theta) : \text{Information matrix}$$

* MLE 특징

① as $n \rightarrow \infty$

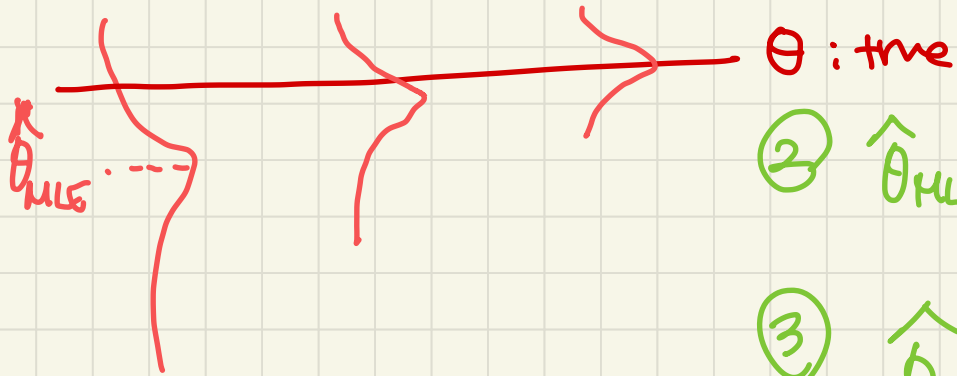
$$\hat{\theta} \xrightarrow{P} \theta$$

convergence in prob.

$n=10$

$n=100$

$n=1000$



② $\hat{\theta}_{MLE} \xrightarrow{a} \text{Normal}$

③ $\hat{\theta}_{MLE} \xrightarrow{a} N(\theta, I(\theta)^{-1})$
 (Cramer Rao Lower bound)
 C.R.L.B
 불확실성의
 하한선